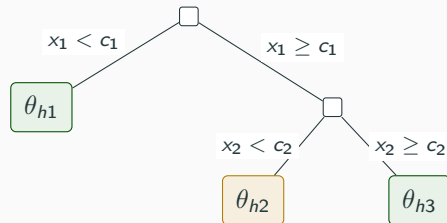


# A shared surface with local discrepancies

**RWD controls:**  $f(x)$       **RCT controls:**  $f(x) + g(x)$

One tree in  $g$  (illustrative splits)



Each profile reaches one leaf per tree.  
Add those contributions:  $g(x) = \sum_{h=1}^{H_g} \theta_{h, \ell_h(x)}$ .

**$f$ : the external response mean**

Fitted jointly to both control sources.

**Spike:**  $z_{h\ell} = 1$

$\theta_{h\ell} \sim N(0, \tau_0^2)$

Draws the RCT mean toward  $f(x)$ .

**Slab:**  $z_{h\ell} = 0$

$\theta_{h\ell} \sim N(0, \tau_1^2), \quad \tau_1^2 > \tau_0^2$

Allows departures from  $f(x)$ .

$z_{h\ell} \sim \text{Bernoulli}(w)$ . Leaf indicators vary;  
 $w, \tau_0^2, \tau_1^2$  are shared.